

Documentation file “_u_init.out”

This document contains the documentation for the chemical data output file ending in “_u_init.out” produced by the code REMIX. The file is written by the Chemistry class through the subroutine `write_u_init`. It reports the initial aqueous chemistry of the medium in terms of aqueous components and variable activity species, for both the domain (target) waters and the external (boundary/recharge) waters.

We describe the output file with a denitrification example: a system with 7 aqueous components built from the primary species, a domain of 10 target waters and two external waters named “inflow” and “initial”. A representative file (“denit_u_init.out”) can be found alongside the simulation output.

Output file: “_u_init.out”

This file contains the initial aqueous component and species concentrations of every water in the domain and of the external waters. It is written by the Chemistry class. All values are expressed in the aqueous component basis of the resident (initial) water type. The numeric values are written in scientific notation (format ES15.5); in every concentration block rows are components or species and columns are waters.

- **Block 1:** Number of aqueous components
 - Number of aqueous components: *num_comps*
 - *num_comps*: number of aqueous components of the resident (initial) water type basis, in which all the values of the file are expressed
- **Block 2:** Aqueous variable activity species (rows 1 to *num_prim*: primary; rows *num_prim*+1 to *num_sp*: secondary)
 - **Line 2a** (repeated once per species): *j: sp_name*
 - *j*: index of the species
 - *sp_name*: name of the variable activity species (same order used in the concentration rows; the species of every water are reordered to this resident order)
- **Block 3:** Aqueous components
 - **Line 3a** (repeated *num_comps* times): *uJ = combination*
 - *combination*: linear combination of variable activity species defined by row *J* of the component matrix *comp_mat_aq* of the resident water type (zero coefficients are omitted; unit coefficients are written without the coefficient)
- **Block 4:** Aqueous variable activity species concentrations in the domain (rows = species, cols = target waters)
 - **Line 4a** (one row per species, format ES15.5): *c(species, :)* (real)
 - Species concentrations of every target water, one column per target water
- **Block 5:** Aqueous component concentrations in the domain (rows = components, cols = target waters)

- **Line 5a** (one row per component, format ES15.5): $u(comp, :)$ (real)
- Component concentrations of every target water, one column per target water (computed as $u = comp_mat_aq \cdot c$)
- **Block 6:** Aqueous variable activity species concentrations of external waters (rows = species, cols = external waters)
 - **Line 6a:** *name* (string)
 - Name of each external water (column header)
 - **Line 6b** (one row per species, format ES15.5): $c(species, :)$ (real)
 - Species concentrations of the external waters
- **Block 7:** Aqueous component concentrations of external waters (rows = components, cols = external waters)
 - **Line 7a:** *name* (string)
 - Name of each external (boundary) water (column header), e.g. inflow, initial
 - **Line 7b** (one row per component, format ES15.5): $u(comp, :)$ (real)
 - Component concentrations of the external waters

End of file

Number of aqueous components: 7

Aqueous variable activity species (rows 1 to 7: primary; rows 8 to 8: secondary):

1: h+
2: cl-
3: no3-
4: n2(aq)(p)
5: ch2o(aq)
6: hco3-
7: o2(aq)
8: co3-2

Aqueous components:

u1 = h+ - co3-2
u2 = cl-
u3 = no3-
u4 = n2(aq)(p)
u5 = ch2o(aq)
u6 = hco3- + co3-2
u7 = o2(aq)

Aqueous variable activity species concentrations in the domain (rows = species, cols = target waters):

1.00000E-07	1.00000E-07	1.00000E-07	1.00000E-07	1.00000E-07	1.00000E-07	1.00000E-07	1.00000E-07	1.00000E-07	1.00000E-07	1.00000E-07
1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16
1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05
1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16
1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05
1.00000E-03	1.00000E-03	1.00000E-03	1.00000E-03	1.00000E-03	1.00000E-03	1.00000E-03	1.00000E-03	1.00000E-03	1.00000E-03	1.00000E-03
1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06
4.69029E-07	4.69029E-07	4.69029E-07	4.69029E-07	4.69029E-07	4.69029E-07	4.69029E-07	4.69029E-07	4.69029E-07	4.69029E-07	4.69029E-07

Aqueous component concentrations in the domain (rows = components, cols = target waters):

-3.69029E-07	-3.69029E-07	-3.69029E-07	-3.69029E-07	-3.69029E-07	-3.69029E-07	-3.69029E-07	-3.69029E-07	-3.69029E-07	-3.69029E-07	-3.69029E-07
1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16
1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05
1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16	1.00000E-16
1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05	1.00000E-05
1.00047E-03	1.00047E-03	1.00047E-03	1.00047E-03	1.00047E-03	1.00047E-03	1.00047E-03	1.00047E-03	1.00047E-03	1.00047E-03	1.00047E-03
1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06	1.00000E-06

Aqueous variable activity species concentrations of external waters (rows = species, cols = external waters):

inflow	initial
1.00000E-08	1.00000E-07
1.00000E-03	1.00000E-16
1.00000E-03	1.00000E-05
1.00000E-16	1.00000E-16
2.00000E-03	1.00000E-05
5.37611E-04	1.00000E-03
2.52947E-04	1.00000E-06
2.52155E-06	4.69029E-07

Aqueous component concentrations of external waters (rows = components, cols = external waters):

inflow	initial
-2.51155E-06	-3.69029E-07
1.00000E-03	1.00000E-16
1.00000E-03	1.00000E-05
1.00000E-16	1.00000E-16
2.00000E-03	1.00000E-05
5.40132E-04	1.00047E-03
2.52947E-04	1.00000E-06